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INKJET PRINTING APPARATUS

Abstract

This inkjet printing apparatus includes an ink circulation path in which a circulation pump that circulates ink and a heater that heats circulating ink are interposed. A control unit can sequentially perform a step (a) of, upon receipt of a signal regarding a command for stopping driving the circulation pump, stopping driving the heater, and determining a circulation duration for which the circulation pump is kept operating and ink circulation in the circulation path is continued, on the basis of a duty value of the heater for a predetermined time period before the time of receipt of the signal and a flow rate of ink in the circulation path for the predetermined time period, and a step (b) of stopping driving the circulation pump after the elapse of the circulation duration.

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Background/Summary

RELATED APPLICATIONS

[0001] This application claims the benefit of Japanese application No. 2024-029380, filed on 29 Feb. 2024, the disclosure of which is incorporated by reference herein.

BACKGROUND OF THE INVENTION

Field of the Invention

[0002] The present invention relates to an inkjet printing apparatus that discharges ink onto a printing medium such as paper to perform printing.

Description of the Background Art

[0003] Conventionally, an inkjet printing apparatus includes an ink circulation path through which ink is supplied to a discharge head configured to discharge ink onto a printing medium, and ink being left undischarged in the discharge head is collected and again supplied to the discharge head, in some cases. An inkjet printing apparatus including such an ink circulation path as mentioned above is described in, for example, Japanese Patent Application Laid-Open No. 2023-053103.

[0004] In an inkjet recording apparatus (1) of Japanese Patent Application Laid-Open No. 2023-053103, a circulation-type ink supply system is adopted. In the inkjet recording apparatus (1), an ink supply unit (15) is provided at some midpoint in a passage connecting a recording head (8) for discharging ink and an ink tank unit (14) in which ink supplied to the recording head (8) is stored. The ink supply unit (15) adjusts a pressure of ink supplied to the recording head (8) and a flow rate of ink collected from the recording head (8) to an appropriate range (paragraph [0016]). Further, a sub-heater not shown that heats ink in the recording head (8) to adjust a temperature is provided (paragraph [0088]).

[0005] In the inkjet recording apparatus (1) of Japanese Patent Application Laid-Open No. 2023-053103, ink circulation starts when a recording operation is performed, and ink circulation stops when a recording operation ends. By ink circulation, ink that begins to be thickened in the vicinity of a discharge port (1006) of the recording head (8) is diffused in the passage. Thus, it is possible to prevent thickened ink from accumulating in the vicinity of the discharge port (1006). Consequently, it is possible to prevent ink from failing to be properly discharged due to adhesion of thickened ink in the vicinity of the discharge port (1006) (paragraphs [0048], [0049], and [0063]).

[0006] Here, in a case in which a recording operation is completed in a short time period, ink circulation also is over in a short time period commensurately with a recording time period. In this case, thickening of ink in the vicinity of the discharge port (1006) cannot be satisfactorily eliminated (paragraph [0050]). Then, in the inkjet recording apparatus (1) of Japanese Patent Application Laid-Open No. 2023-053103, in a case in which a recording operation is completed in a short time period, a process of further continuing circulation for a predetermined time period is performed in addition to ink circulation performed commensurately with the recording operation. More specifically, it is determined whether an elapsed time period (T_c) from a starting time (T_s) of circulation is shorter than a preset value (T_{min}). When the elapsed time period (T_c) is shorter than the preset value (T_{min}), circulation is continued for a predetermined time period (T_{add}) also after the end of the recording operation. Note that the preset value (T_{min}) and the value of the time period (T_{add}) for which circulation is continued can be appropriately set. Thus, thickened ink in the vicinity of the discharge port (1006) is diffused in the passage, so that thickening of ink in the vicinity of the discharge port (1006) can be eliminated (paragraphs and) [0061]).

[0007] However, in Japanese Patent Application Laid-Open No. 2023-053103 described above, the time period for which circulation should be continued in order to eliminate thickening of ink is highly likely to depend largely on a driving condition of the sub-heater that heats ink and a flow rate of ink circulating through the passage. Hence, to set a time period for which circulation is

continued without consideration of those factors would cause a fear that the time period might be more lengthy than necessary or be insufficient. Further, there are many points in time when ink circulation is required to be stopped, other than after the end of a recording operation.

SUMMARY OF THE INVENTION

[0008] It is an object of the present invention to provide a technology that makes it possible to determine a time period for which ink circulation should be continued with consideration given to a driving condition of a heater and a flow rate of circulating ink before the time of receipt of a command for stopping ink circulation in a case in which such a command is received.

[0009] To solve the above-described problem, the first invention of the present application is directed to an inkjet printing apparatus that discharges ink onto a printing medium to perform printing, and includes an ink circulation path, a circulation pump, a heater, an input unit, and a control unit. The circulation path includes a discharge head configured to discharge ink, a supply tank in which ink supplied to the discharge head is stored, a collecting tank in which ink collected from the discharge head is stored, and a feedback pipe connecting the collecting tank and the supply tank. The circulation pump is interposed in the feedback pipe and is configured to deliver ink from the collecting tank to the supply tank via the feedback pipe. The heater is interposed in the feedback pipe and is configured to heat ink flowing from the collecting tank to the supply tank. The input unit is configured to receive a command for stopping driving the circulation pump. The control unit is electrically connected to each of the circulation pump, the heater, and the input unit. The control unit is capable of sequentially performing a step (a) of, upon receipt of a signal regarding the command from the input unit, stopping driving the heater, and determining a circulation duration for which the circulation pump is kept operating and ink circulation in the circulation path is continued from a time of receipt of the signal, on the basis of a duty value of the heater for a predetermined time period before the time of receipt of the signal and a flow rate of ink in the circulation path for the predetermined time period, and a step (b) of stopping driving the circulation pump after elapse of the circulation duration from the time of receipt of the signal.

[0010] The second invention of the present application is directed to the inkjet printing apparatus according to the first invention, wherein the circulation duration is inversely proportional to the flow rate of ink in the circulation path for the predetermined time period.

[0011] The third invention of the present application is directed to the inkjet printing apparatus according to the first or second invention, wherein the circulation duration is proportional to the duty value of the heater for the predetermined time period.

[0012] According to the first to third inventions of the present application, in a case in which a command for stopping ink circulation is received, the heater is stopped operating while ink circulation is continued. Thus, the temperature of the heater can be reduced. This can suppress a change in quality or the like of ink left in the heater. Further, a time period for which ink circulation is continued is determined by a duty value of the heater and a flow rate of circulating ink, and therefore the time period can be minimized.

[0013] Especially, according to the second invention of the present application, the higher the flow rate of ink is, the more efficiently the temperature of the heater can be reduced in circulating ink. This can shorten the circulation duration.

[0014] These and other objects, features, aspects and advantages of the present invention will become more apparent from the following detailed description of the present invention when taken in conjunction with the accompanying drawings.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0015] FIG. 1 is a view conceptually showing a configuration of an inkjet printing apparatus;

[0016] FIG. 2 is a view conceptually showing configurations of an ink supply unit and a discharge head;

[0017] FIG. 3 is a block diagram showing connection between a control unit and each component of the inkjet printing apparatus;

[0018] FIG. 4 is a flowchart showing a procedure for conveyance of continuous paper, printing on the continuous paper, ink circulation, and stop of ink circulation; and

[0019] FIG. 5 is a view showing a relationship between a temperature of ink passing through a heater and a duty value of the heater, plotted against the temperature.

DESCRIPTION OF THE PREFERRED EMBODIMENTS

[0020] Hereinafter, a preferred embodiment of the present invention will be described with reference to the drawings. Note that components described in the preferred embodiment are mere examples and are not intended to limit the scope of the present invention to those only. In the drawings, for the purpose of easier understanding, the dimensions or the number of respective components are overstated or understated in some portions of illustration, as necessary.

<1. Configuration of Inkjet Printing Apparatus>

[0021] FIG. 1 is a view conceptually showing a configuration of an inkjet printing apparatus **1** according to one preferred embodiment of the present invention. The inkjet printing apparatus **1** is an inkjet printing machine that discharges droplets of water-based ink onto continuous paper **10** being in a shape of a long strip from a plurality of discharge heads **35** while conveying the continuous paper **10**, to record characters or images on a surface of the continuous paper **10**. Note that the continuous paper **10** in a shape of a long strip is just one example of a printing medium. The printing medium may be a cut sheet, a plastic film, cardboard, metal foil, a glass material, or the like. In other words, the inkjet printing apparatus **1** may be any apparatus that can discharge ink onto a printing medium to perform printing. As shown in FIG. 1, the inkjet printing apparatus **1** includes a conveyor unit **2**, a printing unit **3**, a control unit **9**, and an input unit **11**.

[0022] The conveyor unit **2** is a mechanism configured to convey the continuous paper **10** along a predetermined conveying path in a conveying direction extending along a length direction of the continuous paper **10**. The continuous paper **10** is stretched over a plurality of conveyor rollers **12**. The continuous paper **10** is conveyed along a conveying path formed by the plurality of conveyor rollers **12**. Each of the conveyor rollers **12** rotates about an axis extending in a direction perpendicular to the conveying direction, to thereby guide the continuous paper **10** to the downstream side in the conveying path. Further, the continuous paper **10** is under tension in the conveying direction. This reduces slack or wrinkles in the continuous paper **10** during conveying.

[0023] The printing unit **3** includes a plurality of discharge heads **35** and a plurality of ink supply units **4**. In the present preferred embodiment, the printing unit **3** includes four discharge heads **35** and four ink supply units **4**. The four discharge heads **35** have substantially the same configuration with each other. Further, the four ink supply units **4** have substantially the same configuration with each other.

[0024] The four discharge heads **35** are arranged while being spaced from each other along the conveying direction. Each of the four discharge heads **35** discharges ink droplets onto a surface (upper surface) of the continuous paper **10** from nozzles **83** (refer to FIG. 2 described later). In the present preferred embodiment, the four discharge heads **35** discharge ink of different colors, respectively, to thereby each record a monochromatic image on the surface (upper surface) of the continuous paper **10**. In the present preferred embodiment, for example, the four discharge heads **35** discharge cyan ink, magenta ink, yellow ink, and black ink, respectively. Then, the four monochromatic images are superimposed, so that a multicolor image is formed on the upper surface of the continuous paper **10**.

[0025] FIG. 2 is a view conceptually showing a configuration of one ink supply unit **4** and a configuration of one discharge head **35**. In the present preferred embodiment, each of the discharge heads **35** includes a plurality of heads **80**. In the present preferred embodiment, each of the

discharge heads **35** includes five heads **80**. The five heads **80** have substantially the same configuration with each other. Hence, in FIG. 2, only one of the five heads **80** is shown in detail, and the other four heads **80** are shown in a simplified manner. As shown in FIG. 2, each of the five heads **80** includes a casing **81**, an internal tank **82**, and a plurality of nozzles **83**.

[0026] The casing **81** forms an outer frame of the head **80**. The internal tank **82** is provided in the casing **81**, and ink can be temporarily stored therein. The plurality of nozzles **83** are arranged while being equally spaced from each other along the conveying direction and a width direction of the continuous paper **10** in a lower portion of the casing **81**. Each of the plurality of nozzles **83** communicates with the internal tank **82**. Further, each of the plurality of nozzles **83** includes a plurality of piezoelectric elements **831** serving as pressure generation elements, an ink chamber **832**, and a discharge port **830**. The ink chamber **832** communicates with the internal tank **82**.

[0027] During discharge of ink, ink flows down from the internal tank **82** to the ink chamber **832**. Then, under the control of the piezoelectric elements **831**, ink in the ink chamber **832** is pressurized, and thus is discharged in the form of liquid droplets from the discharge port **830**. Alternatively, the nozzle **83** may be a so-called thermal nozzle in which ink in the ink chamber **832** is heated to generate bubbles and thus is pressurized.

[0028] Next, the ink supply unit **4** is described. The ink supply unit **4** is a device configured to supply ink to the discharge heads **35** while circulating a part of ink. As described above, the inkjet printing apparatus **1** of the present preferred embodiment includes four ink supply units **4**. The four ink supply units **4** have substantially the same configuration with each other, and hence only a configuration of one ink supply unit **4** is described below.

[0029] As shown in FIG. 2, each of the ink supply units **4** includes a supply tank **51**, a collecting tank **52**, a supply-side manifold **61**, a plurality of supply-side narrow pipes **62**, a plurality of collecting-side narrow pipes **63**, a collecting-side manifold **64**, a feedback pipe **65**, a circulation pump **71**, a plurality of supply-side on-off valves **73**, a plurality of head outlet-side on-off valves **74**, a feedback-side on-off valve **75**, a first heater **76**, a second heater **77**, a first temperature sensor **84**, a second temperature sensor **85**, a third temperature sensor **86**, a filter **87**, and a deaeration unit **88**. In the present preferred embodiment, each of the ink supply units **4** includes five supply-side narrow pipes **62**, five collecting-side narrow pipes **63**, five supply-side on-off valves **73**, and five head outlet-side on-off valves **74**.

[0030] The supply tank **51** is a container for temporally storing ink to be supplied to the discharge heads **35**. In the supply tank **51**, an internal chamber **510** in which ink can be temporally stored is provided. Meanwhile, in the supply tank **51**, a liquid-level sensor for detecting a liquid level of ink stored in the internal chamber **510** of the supply tank **51** may be provided.

[0031] The supply-side manifold **61** and the five supply-side narrow pipes **62** are pipes connecting the supply tank **51** and the five heads **80** included in one discharge head **35**. The supply-side manifold **61** is a wide pipe having an upstream end that is connected so as to communicate with the internal chamber **510** of the supply tank **51**. Each of the five supply-side narrow pipes **62** is a narrow pipe branching from the supply-side manifold **61**. Each of the five supply-side narrow pipes **62** has an upstream end communicating with an internal passage of the supply-side manifold **61**, and has a downstream end that is connected so as to communicate with the internal tank **82** of one head **80**.

[0032] Further, in the present preferred embodiment, the supply-side on-off valve **73** is interposed in each of the supply-side narrow pipes **62**. For the supply-side on-off valve **73**, for example, a solenoid valve that is opened and closed under the control of the control unit **9** is used.

Alternatively, for the supply-side on-off valve **73**, an on-off valve that is manually opened and closed may be used. While the supply-side on-off valve **73** is closed, an internal passage of the supply-side narrow pipe **62** is blocked from communicating. That is, while the supply-side on-off valve **73** is closed, ink flow from the supply tank **51** to the head **80** is interrupted. Meanwhile, while the supply-side on-off valve **73** is opened, the internal passage of the supply-side narrow pipe

62 is allowed to communicate. Note that the supply-side on-off valve **73** is not necessarily required to be provided. Further, a filter or the like may be further interposed in the supply-side manifold **61** or each of the five supply-side narrow pipes **62**.

[0033] The five collecting-side narrow pipes **63** and the collecting-side manifold **64** are pipes connecting the five heads **80** included in one discharge head **35** and the collecting tank **52**. Each of the five collecting-side narrow pipes **63** is a narrow pipe branching from the collecting-side manifold **64**. Each of the five collecting-side narrow pipes **63** has an upstream end that is connected so as to communicate with the internal tank **82** of one head **80**, and has a downstream end that is connected so as to communicate with an internal passage of the collecting-side manifold **64**. The collecting-side manifold **64** is a wide pipe having a downstream end that is connected so as to communicate with the internal chamber **520** of the collecting tank **52** described later.

[0034] Further, in the present preferred embodiment, the head outlet-side on-off valve **74** is interposed in each of the collecting-side narrow pipes **63**. For the head outlet-side on-off valve **74**, for example, a solenoid valve that is opened and closed under the control of the control unit **9** is used. Alternatively, for the head outlet-side on-off valve **74**, an on-off valve that is manually opened and closed may be used. While the head outlet-side on-off valve **74** is closed, an internal passage of the collecting-side narrow pipe **63** is blocked from communicating. That is, while the head outlet-side on-off valve **74** is closed, ink flow from the head **80** to the collecting tank **52** is interrupted. Meanwhile, while the head outlet-side on-off valve **74** is opened, the internal passage of the collecting-side narrow pipe **63** is allowed to communicate. Note that the head outlet-side on-off valve **74** is not necessarily required to be provided. Further, a filter or the like may be further interposed in each of the five collecting-side narrow pipes **63** or the collecting-side manifold **64**.

[0035] The collecting tank **52** is a container for temporally storing ink collected from the discharge heads **35**. In the collecting tank **52**, the internal chamber **520** in which ink can be temporally stored is provided. Meanwhile, in the collecting tank **52**, a liquid-level sensor for detecting a liquid level of ink stored in the internal chamber **520** of the collecting tank **52** may be provided.

[0036] Further, as shown in FIG. 2, the supply tank **51** is connected to a pressurization mechanism **515**. The pressurization mechanism **515** pressurizes the inside of the supply tank **51**, to regulate a pressure of the internal chamber **510** of the supply tank **51** to a positive pressure. That is, the pressurization mechanism **515** pressurizes the inside of the supply tank **51**, to regulate a pressure of the internal chamber **510** of the supply tank **51** to a pressure higher than the atmospheric pressure. The pressurization mechanism **515** includes, for example, a compressor, a pressurization buffer tank, a pressure regulation mechanism (regulator), and the like. Meanwhile, the collecting tank **52** is connected to a decompression mechanism **524**. The decompression mechanism **524** decompresses the inside of the collecting tank **52**, to regulate a pressure of the internal chamber **520** of the collecting tank **52** to a negative pressure. That is, the decompression mechanism **524** decompresses the inside of the collecting tank **52**, to regulate a pressure of the internal chamber **520** of the collecting tank **52** to a pressure lower than the atmospheric pressure. The decompression mechanism **524** includes, for example, a vacuum pump, a decompression buffer tank, a pressure regulation mechanism (regulator), and the like.

[0037] The pressurization mechanism **515** and the decompression mechanism **524** are configured such that the operations thereof can be controlled by the control unit **9**. When the pressurization mechanism **515** and the decompression mechanism **524** are driven, there is generated a pressure difference between the internal chamber **510** of the supply tank **51** and the internal chamber **520** of the collecting tank **52**. As a result, ink stored in the supply tank **51** can be supplied to the discharge heads **35**, and further, ink remaining in the discharge heads **35** can be collected into the collecting tank **52**. Note that the ink remaining in the discharge heads **35** is ink being left undischarged in the discharge heads **35**.

[0038] Note that the pressurization mechanism **515** and the decompression mechanism **524** are not necessarily, required to make a pressure of the internal chamber **510** of the supply tank **51** positive

and required to make a pressure of the internal chamber 520 of the collecting tank 52 negative, as long as those pressures can be adjusted such that the pressure of the internal chamber 510 of the supply tank 51 is higher than the pressure of the internal chamber 520 of the collecting tank 52. For example, the pressurization mechanism 515 may adjust a pressure of the internal chamber 510 of the supply tank 51 to a pressure equal to the atmospheric pressure, and the decompression mechanism 524 may adjust a pressure of the internal chamber 520 of the collecting tank 52 to a negative pressure. That is, the pressurization mechanism 515 may adjust a pressure of the internal chamber 510 of the supply tank 51 to a pressure equal to the atmospheric pressure, and the decompression mechanism 524 may adjust a pressure of the internal chamber 520 of the collecting tank 52 to a pressure lower than the atmospheric pressure.

[0039] The feedback pipe 65 is a pipe connecting the internal chamber 520 of the collecting tank 52 and the internal chamber 510 of the supply tank 51 in such a manner as to allow the internal chambers to communicate with each other. In other words, the feedback pipe 65 connects the collecting tank 52 and the supply tank 51. As shown in FIG. 2, an internal passage of the feedback pipe 65 has an upstream end that is connected so as to communicate with the internal chamber 520 of the collecting tank 52. Further, the internal passage of the feedback pipe 65 has a downstream end that is connected so as to communicate with the internal chamber 510 of the supply tank 51.

[0040] With the above-described configuration, there is formed an ink circulation path 41 that starts from the supply tank 51, extends through the supply-side manifold 61, the supply-side narrow pipes 62, the internal tanks 82 of the discharge heads 35, the collecting-side narrow pipes 63, the collecting-side manifold 64, the collecting tank 52, and the feedback pipe 65, and returns back to the supply tank 51. That is, the ink circulation path 41 includes the supply tank 51, the discharge heads 35, the collecting tank 52, and the feedback pipe 65. Further, in the feedback pipe 65, the circulation pump 71, the feedback-side on-off valve 75, the first heater 76, the second heater 77, the first temperature sensor 84, the second temperature sensor 85, the third temperature sensor 86, the filter 87, and the deaeration unit 88 are interposed.

[0041] The circulation pump 71 is a device configured to perform a pumping operation of delivering ink from the collecting tank 52 to the supply tank 51 via the feedback pipe 65. The circulation pump 71 generates ink flow from the collecting tank 52 to the supply tank 51 in the internal passage of the feedback pipe 65 in response to an operation signal from the control unit 9. For the circulation pump 71 of the present preferred embodiment, for example, a pump in which foreign matters such as dust are unlikely to be generated during driving, such as a diaphragm pump, is used. The circulation pump 71 applies a pressure to ink in the circulation pump 71 by reciprocation of a piston therein. Then, the circulation pump 71 discharges ink from an outlet communicating with the internal passage of the feedback pipe 65. Further, the circulation pump 71 is electrically connected to the control unit 9. The circulation pump 71 outputs data regarding a value calculated from “a load factor×a ratio of a time period of an ON state to a unit time period”, to the control unit 9. Note that the unit time period is, for example, one second. Hereinafter, the value calculated from “a load factor×a ratio of a time period of an ON state to a unit time period” will be referred to as a “pump duty value” in order to be distinguished from duty values of the first and second heaters 76 and 77 described later.

[0042] The feedback-side on-off valve 75 is interposed on the downstream side of the circulation pump 71 with respect to an ink delivery direction and on the upstream side of the first heater 76 and the second heater 77 with respect to the ink delivery direction in the feedback pipe 65. For the feedback-side on-off valve 75, for example, a solenoid valve that is opened and closed under the control of the control unit 9 is used. Alternatively, for the feedback-side on-off valve 75, an on-off valve that is manually opened and closed may be used. While the feedback-side on-off valve 75 is closed, the internal passage of the feedback pipe 65 is blocked from communicating. That is, while the feedback-side on-off valve 75 is closed, ink flow from the collecting tank 52 to the supply tank 51 and backflow of ink from the supply tank 51 to the collecting tank 52 are prevented. Meanwhile,

while the feedback-side on-off valve **75** is opened, the internal passage of the feedback pipe **65** is allowed to communicate. Thus, the feedback-side on-off valve **75** allows or interrupts ink flow from the collecting tank **52** to the supply tank **51**.

[0043] The first heater **76** is a device configured to heat ink delivered through the internal passage of the feedback pipe **65**. The first heater **76** heats ink flowing from the collecting tank **52** to the supply tank **51**. The first heater **76** is positioned between the circulation pump **71** and the supply tank **51** in the feedback pipe **65**. The first heater **76** includes a heating element formed of a carbon heater or the like and is connected to a power supply via an ON/OFF circuit not shown. Then, the first heater **76** can heat ink by generating heat in an ON state in which power is turned on. Further, the first heater **76** is controlled such that, for example, a temperature of ink passing through the first heater **76** is adjusted toward 30° C. as a desired temperature, by switching between an ON state and an OFF state.

[0044] The second heater **77** is a device configured to heat ink delivered through the internal passage of the feedback pipe **65**. The second heater **77** heats ink flowing from the collecting tank **52** to the supply tank **51**. The second heater **77** is positioned between the first heater **76** and the supply tank **51** in the feedback pipe **65**. That is, the second heater **77** is positioned on the downstream side of the first heater **76** with respect to the ink delivery direction. The second heater **77** includes a heating element formed of a carbon heater or the like and is connected to the power supply via the ON/OFF circuit not shown. Then, the second heater **77** can heat ink by generating heat in an ON state in which power is turned on. Further, the second heater **77** is controlled such that, for example, a temperature of ink passing through the second heater **77** is adjusted toward 35° C. as a desired temperature, by switching between an ON state and an OFF state.

[0045] Meanwhile, each of the first heater **76** and the second heater **77** is electrically connected to the control unit **9**. The first heater **76** and the second heater **77** each output data regarding a ratio of a time period of an ON state to a unit time period, to the control unit **9**. Note that the unit time period is, for example, one second. Hereinafter, the ratio of a time period of an ON state to a unit time period will be referred to as a “duty value”. Note that, in the present invention, the number of heaters interposed between the circulation pump **71** and the supply tank **51** in the feedback pipe **65** may be one or three or larger. That is, any heater that is interposed between the circulation pump **71** and the supply tank **51** in the feedback pipe **65** and heats ink flowing from the collecting tank **52** to the supply tank **51** can be used.

[0046] The first temperature sensor **84** is interposed on the downstream side of the feedback-side on-off valve **75** with respect to the ink delivery direction and on the upstream side of the first heater **76** with respect to the ink delivery direction in the feedback pipe **65**. The first temperature sensor **84** detects a temperature of ink flowing into the first heater **76**. Further, the first temperature sensor **84** is electrically connected to the control unit **9**. The first temperature sensor **84** outputs data regarding a result of detection of an ink temperature, to the control unit **9**.

[0047] The second temperature sensor **85** is interposed on the downstream side of the first heater **76** with respect to the ink delivery direction and on the upstream side of the second heater **77** with respect to the ink delivery direction in the feedback pipe **65**. The second temperature sensor **85** detects a temperature of ink flowing out of the first heater **76**. Further, the second temperature sensor **85** is electrically connected to the control unit **9**. The second temperature sensor **85** outputs data regarding a result of detection of an ink temperature, to the control unit **9**.

[0048] The third temperature sensor **86** is interposed on the downstream side of the second heater **77** with respect to the ink delivery direction and on the upstream side of the supply tank **51** with respect to the ink delivery direction in the feedback pipe **65**. The third temperature sensor **86** detects a temperature of ink flowing out of the second heater **77**. Further, the third temperature sensor **86** is electrically connected to the control unit **9**. The third temperature sensor **86** outputs data regarding a result of detection of an ink temperature, to the control unit **9**. Note that the first temperature sensor **84**, the second temperature sensor **85**, and the third temperature sensor **86**

described above are not necessarily required to be provided.

[0049] The filter **87** is interposed on the downstream side of the second heater **77** with respect to the ink delivery direction and on the upstream side of the supply tank **51** with respect to the ink delivery direction in the feedback pipe **65**. The filter **87** filters ink delivered through the internal passage of the feedback pipe **65**, to remove foreign matters included in the ink.

[0050] The deaeration unit **88** is interposed on the downstream side of the filter **87** with respect to the ink delivery direction and on the upstream side of the supply tank **51** with respect to the ink delivery direction in the feedback pipe **65**. The deaeration unit **88** of the present preferred embodiment is a so-called hollow-fiber membrane deaeration module. The deaeration unit **88** removes bubbles in ink delivered through the internal passage of the feedback pipe **65**.

[0051] Next, the control unit **9** is described. The control unit **9** is an information processing device configured to control each component of the inkjet printing apparatus **1**. FIG. **3** is a block diagram showing connection between the control unit **9** and each component of the inkjet printing apparatus **1**. As conceptually shown in FIG. **3**, the control unit **9** includes a processor **91** such as a CPU, a memory **92** such as a RAM, and a storage unit **93** such as a hard disk drive. In the storage unit **93**, a computer program **9P** for performing conveyance of the continuous paper **10**, printing on the continuous paper **10**, and ink circulation, and for stopping ink circulation in response to a command described later, is stored.

[0052] Further, as shown in FIG. **3**, the control unit **9** is electrically connected to the conveyor unit **2** and the four discharge heads **35** of the printing unit **3**, and is further electrically connected to the circulation pump **71**, the five supply-side on-off valves **73**, the five head outlet-side on-off valves **74**, the feedback-side on-off valve **75**, the first heater **76**, the second heater **77**, the respective temperature sensors **84**, **85**, and **86**, the pressurization mechanism **515**, and the decompression mechanism **524** of each of the four ink supply units **4** of the printing unit **3**, such that the control unit **9** can conduct communication to/from the above-described components. The control unit **9** controls operations of those components in accordance with the computer program **9P**.

[0053] Moreover, as shown in FIG. **3**, the control unit **9** is electrically connected to the input unit **11** such that the control unit **9** can conduct communication to/from the input unit **11**. The input unit **11** is a device configured to receive a command for stopping driving the circulation pump **71**. The input unit **11** includes an input interface such as a touch panel, for example. An operator inputs a command for, for example, stopping driving the circulation pump **71** to the input unit **11** via the input interface. Then, upon receipt of the command, the input unit **11** outputs a signal regarding the command to the control unit **9**. Note that the input unit **11** may be formed integrally with the control unit **9** or another unit. The control unit **9** controls operations of the above-described components, to perform conveyance of the continuous paper **10**, printing on the continuous paper **10**, and ink circulation in the circulation path **41**, and further, stop ink circulation with a time lag in response to the signal as described later.

<2. Procedure for Conveyance of Continuous Paper, Printing, Ink Circulation, and Stop of Ink Circulation>

[0054] Next, description is given about a procedure for conveyance of the continuous paper **10**, printing on the continuous paper **10**, ink circulation, and stop of ink circulation that are performed in the inkjet printing apparatus **1**. FIG. **4** is a flowchart showing a procedure for conveyance of the continuous paper **10**, printing on the continuous paper **10**, ink circulation, and stop of ink circulation.

[0055] In performing conveyance of the continuous paper **10**, printing on the continuous paper **10**, and ink circulation, the control unit **9** first causes the conveyor unit **2** to operate, to convey the continuous paper **10** along the predetermined conveying path in the length direction. Then, the control unit **9** controls the plurality of nozzles **83** of each of the four discharge heads **35** such that ink droplets are discharged onto the surface of the continuous paper **10**. Thus, an image is recorded on the surface of the continuous paper **10** (step **S1**).

[0056] Meanwhile, as advance preparation for conveyance of the continuous paper **10**, printing on the continuous paper **10**, and ink circulation, a sufficient amount of ink is stored in the internal chamber **510** of the supply tank **51**. Further, the control unit **9** opens the five supply-side on-off valves **73**, the five head outlet-side on-off valves **74**, and the feedback-side on-off valve **75**.

[0057] Then, the control unit **9** drives the circulation pump **71**, the first heater **76**, the second heater **77**, the pressurization mechanism **515**, and the decompression mechanism **524** of each of the four ink supply units **4**. Further, the first temperature sensor **84**, the second temperature sensor **85**, and the third temperature sensor **86** are turned on the power of them, and thus, those temperature sensors start measuring. That is, the control unit **9** drives the circulation pump **71** to circulate ink through the ink circulation path **41**, while driving the pressurization mechanism **515** and the decompression mechanism **524** to supply ink to the internal tank **82** of each discharge head **35**.

[0058] More specifically, when the pressurization mechanism **515** and the decompression mechanism **524** are driven, a pressure difference is generated between the internal chamber **510** of the supply tank **51** and the internal chamber **520** of the collecting tank **52**. As a result, ink stored in the supply tank **51** can be supplied to the discharge heads **35**, and further, ink remaining in the discharge heads **35** can be collected into the collecting tank **52**. Note that the ink remaining in the discharge heads **35** is ink being left undischarged in the discharge heads **35**. Further, when the circulation pump **71** is driven, there is generated ink flow from the collecting tank **52** to the supply tank **51** in the internal passage of the feedback pipe **65**.

[0059] Meanwhile, a volume of a portion where ink passes in the supply tank **51**, the collecting tank **52**, the supply-side manifold **61**, each of the supply-side narrow pipes **62**, each of the collecting-side narrow pipes **63**, the collecting-side manifold **64**, and the feedback pipe **65** on the ink circulation path **41** is a preset value. Hence, a flow rate of ink circulating through the circulation path **41** can be calculated from a pumping rate of the circulation pump **71** that delivers ink, per unit time period, and a pressure of the internal chamber **510** of the supply tank **51** and a pressure of the internal chamber **520** of the collecting tank **52** that are adjusted by the pressurization mechanism **515** and the decompression mechanism **524**. Note that the pumping rate of the circulation pump **71** per unit time period can be calculated from the above-described “pump duty value”.

[0060] On this basis, the inkjet printing apparatus **1** has a configuration that allows the operator to select a “high-speed circulation mode” or a “low-speed circulation mode” for a flow rate of ink circulating through the circulation path **41**. When the “high-speed circulation mode” is selected, the circulation pump **71** is controlled such that, for example, the pump duty value is “100%”. Further, the pressurization mechanism **515** is controlled such that, for example, a pressure of the internal chamber **510** of the supply tank **51** is “+10 kPa”. Moreover, the decompression mechanism **524** is controlled such that, for example, a pressure of the internal chamber **520** of the collecting tank **52** is “-10 kPa”. As a result of this, a flow rate of ink in the circulation path **41** of the present preferred embodiment is, for example, 800 (ml/min).

[0061] Meanwhile, when the “low-speed circulation mode” is selected, the circulation pump **71** is controlled such that, for example, the pump duty value is “50%”. Further, the pressurization mechanism **515** is controlled such that, for example, a pressure of the internal chamber **510** of the supply tank **51** is “+3 kPa”. Moreover, the decompression mechanism **524** is controlled such that, for example, a pressure of the internal chamber **520** of the collecting tank **52** is “-3 kPa”. As a result of this, a flow rate of ink in the circulation path **41** of the present preferred embodiment is, for example, 300 (ml/min).

[0062] Alternatively, for a flow rate of ink circulating through the circulation path **41**, there may be provided one mode or three or more modes in each of which the pump duty value and the values of pressures of the internal chambers **510** and **520** are different from the above-mentioned values, in addition to the “high-speed circulation mode” and the “low-speed circulation mode”. Further alternatively, a flowmeter may be additionally provided in order to measure a flow rate of ink in the

circulation path **41**.

[0063] Further, by driving the first and second heaters **76** and **77**, it is possible to heat ink delivered through the internal passage of the feedback pipe **65** and then deliver the heated ink to the supply tank **51**. More specifically, the control unit **9** first switches the first heater **76** between an ON state and an OFF state, to control a temperature of ink passing through the first heater **76**, at 30° C., for example. To this end, as described above, the second temperature sensor **85** detects a temperature of ink flowing out of the first heater **76** and outputs data regarding a result of the detection to the control unit **9**. The control unit **9** adjusts a driving amount of the first heater **76** in accordance with the result of the detection of the ink temperature received from the second temperature sensor **85**.

[0064] FIG. **5** is a view showing an example of a ratio of a time period of an ON state to a unit time period in driving each of the first and second heaters **76** and **77**, plotted against a temperature of ink passing through each of the first and second heaters **76**. Note that the unit time period is, for example, one second. In other words, FIG. **5** is a view showing an example of a “duty value” in driving each of the first and second heaters **76** and **77**, plotted against a temperature of ink passing through each of the first and second heaters **76** and **77**. As shown in FIG. **5**, when a temperature of ink passing through the first heater **76** is lower than 27.5° C., the control unit **9** drives the first heater **76** at a duty value of 100%. When the temperature of ink passing through the first heater **76** becomes equal to or higher than 27.5° C., the control unit **9** gradually decreases the duty value. Then, when the temperature of ink becomes equal to or higher than 28° C., the control unit **9** drives the first heater **76** at a duty value of 50% or smaller. Further, when the temperature of ink passing through the first heater **76** becomes equal to or higher than 30° C., the control unit **9** drives the first heater **76** at a duty value of 0%. In other words, when the temperature of ink passing through the first heater **76** becomes equal to or higher than 30° C., the control unit **9** stops driving the first heater **76**. By the above-described control, the temperature of ink passing through the first heater **76** can be adjusted to 30° C.

[0065] Meanwhile, the control unit **9** performs control in which a temperature of ink passing through the second heater **77** becomes equal to, for example, 35° C.” by switching the second heater **77** between an ON state and an OFF state. As described above, the third temperature sensor **86** detects a temperature of ink flowing out of the second heater **77**, and outputs data regarding a result of the detection, to the control unit **9**. The control unit **9** adjusts a driving amount of the second heater **77** in accordance with the result of the detection of the ink temperature received from the third temperature sensor **86**.

[0066] As shown in FIG. **5**, when a temperature of ink passing through the second heater **77** is lower than 28.5° C., the control unit **9** drives the second heater **77** at a duty value of 100%. When the temperature of ink passing through the second heater **77** becomes equal to or higher than 28.5° C., the control unit **9** gradually decreases the duty value. Then, when the temperature of ink becomes equal to or higher than 30° C., the control unit **9** drives the second heater **77** at a duty value of 50% or smaller. Further, when the temperature of ink passing through the second heater **77** becomes equal to or higher than 31.5° C., the control unit **9** drives the second heater **77** at a duty value of 0%. In other words, when the temperature of ink passing through the second heater **77** becomes equal to or higher than 31.5° C., the control unit **9** stops driving the second heater **77**. By the above-described control, the temperature of ink passing through the second heater **77** can be adjusted to 35° C. The adjustment of the temperature of ink passing through the second heater **77** to 35° C. allows the discharge heads **35** to satisfactorily discharge ink afterward.

[0067] Meanwhile, as described above, the first temperature sensor **84** detects a temperature of ink flowing into the first heater **76**, and outputs data regarding a result of the detection, to the control unit **9**. The control unit **9** can find at what temperature ink has originally flowed into the first heater **76**, by referring to the data regarding the result of the detection provided from the first temperature sensor **84**. This serves as a measure to determine whether the first heater **76** and the second heater **77** normally operate. Further, historical data of the driving amounts (duty values) given to the first

heater **76** and the second heater **77** by the control unit **9** is stored in the memory **92** so that the driving amounts can be referred back to at any appropriate time.

[0068] Further, ink flowing through the internal passage of the feedback pipe **65** is caused to pass through the filter **87**, and thus, ink from which small impurities or the like remaining therein have been removed can be delivered to the supply tank **51**. Moreover, ink flowing through the internal passage of the feedback pipe **65** is caused to pass through the deaeration unit **88**, and thus ink having been deaerated can be delivered to the supply tank **51**.

[0069] Here, the circulation pump **71** is stopped operating in some cases such as a case in which switching between the “high-speed circulation mode” and the “low-speed circulation mode” described above is performed and a case in which an operation of wiping out stains on the discharge heads **35** is performed. In such a case, ink circulation in the circulation path **41** is stopped, and thus ink stays in each of the first and second heaters **76** and **77**. Further, power supply to the first and second heaters **76** and **77** is interrupted.

[0070] In this regard, as described above, each of the first and second heaters **76** and **77** includes a heating element formed of a carbon heater or the like. The heating element is kept at a high temperature also after interruption of power supply to the first and second heaters **76** and **77**. Hence, to leave ink circulation stopped for a certain time period would result in overheating ink staying in the first and second heaters **76** and **77**, so that the ink might possibly be changed in quality or deteriorated. Further, to use ink having been changed in quality or deteriorated afterward would cause a fear that ink might not be satisfactorily discharged from the discharge heads **35** or a desired color gamut might not be obtained. For this reason, in the present preferred embodiment, the following process is performed before the circulation pump **71** is stopped operating.

[0071] Specifically, the operator first inputs a command for stopping driving the circulation pump **71** via the input interface of the input unit **11**. Then, upon receipt of the command, the input unit **11** outputs a signal regarding the command to the control unit **9**. The control unit **9** detects whether the signal regarding the command has been received from the input unit **11** (step **S2**). The control unit **9** continues conveyance of the continuous paper **10**, printing on the continuous paper **10**, and ink circulation until it is detected that the signal regarding the command has been received from the input unit **11** (step **S2**: NO).

[0072] On the other hand, when it is detected that the signal regarding the command has been received from the input unit **11** (step **S2**: YES), the control unit **9** refers to each of a duty value of the first heater **76** and a duty value of the second heater **77** for a predetermined time period before it is detected that the signal regarding the command has been received from the input unit **11**. Note that the predetermined time period is, for example, one second. However, the duty values vary in some cases as described above, and thus, the control unit **9** may set a time period of a few minutes as the predetermined time period. In this case, the control unit **9** may refer to a value calculated as an average value per unit time period, the value being obtained by division of an integral value of duty values of each of the first and second heaters **76** and **77** for the predetermined time period, by the predetermined time period.

[0073] Note that, when duty values of the first and second heaters **76** and **77** for the above-described predetermined time period are large, the control unit **9** determines that the heating elements have already generated much heat and the temperatures are high in the first and second heaters **76** and **77**. In such a case, the control unit **9** determines that ink staying in the first and second heaters **76** and **77** are highly likely to be overheated and be changed in quality or deteriorated by immediate stop of ink circulation in the circulation path **41**.

[0074] Further, the control unit **9** refers to whether the “high-speed circulation mode” or the “low-speed circulation mode” has been selected for a flow rate of ink circulating through the circulation path **41** for the above-described predetermined time period. Then, the control unit **9** grasps a flow rate Q_i (ml/min) of ink circulating through the circulation path **41** for the predetermined time period on the basis of the result of referring. For example, when the “high-speed circulation mode”

is selected for the predetermined time period, the flow rate of ink is 800 (ml/min). When the “low-speed circulation mode” is selected for the predetermined time period, the flow rate of ink is 300 (ml/min). Alternatively, the flow rate of ink circulating through the circulation path **41** may be grasped on the basis of a result of measurement provided by a flowmeter additionally provided. [0075] Subsequently, the control unit **9** calculates a circulation duration ΔT (s), by substituting the above-described flow rate Q_i (ml/min) of ink having been grasped and a value $D1$ (%) that is the larger of the respective duty values of the first heater **76** and the second heater **77** for the above-described predetermined time period, into the following expression (1). Then, the control unit **9** continues driving the circulation pump **71** for the circulation duration ΔT (s) to perform ink circulation in the circulation path **41** also after it is detected that the above-described signal has been received (step **S3**). Note that, in the expression (1), “ K_s ” represents a coefficient determined in advance by a length of the ink circulation path **41**, an inner diameter of the feedback pipe **65**, and the like.

[00001] $T(s) = K_s \times D1(\%) \div Q_i(\text{ml} / \text{min})$ Expression(1)

[0076] Then, after the elapse of the circulation duration ΔT (s), the control unit **9** stops driving the circulation pump **71**, to stop ink circulation in the circulation path **41** (step **S4**). That is, the control unit **9** of the present preferred embodiment can sequentially perform a step (a), upon receipt of a signal regarding a command for stopping driving the circulation pump **71** from the input unit **11**, stopping driving the first and second heaters **76** and **77** and determining the circulation duration ΔT (s) for which the circulation pump **71** is kept operating and ink circulation in the circulation path **41** is continued from the time of receipt of the signal, on the basis of the duty values of the first and second heaters **76** and **77** for a predetermined time period before the time of receipt of the signal and the flow rate of ink in the circulation path **41** for the predetermined time period, and a step (b) of stopping driving the circulation pump **71** after the elapse of the circulation duration ΔT (s) from the time of receipt of the signal.

[0077] In this manner, in the present preferred embodiment, in a case in which a command for stopping ink circulation is received, the first and second heaters **76** and **77** are stopped operating, and ink circulation is continued for a predetermined time period. Thus, the temperatures of the first and second heaters **76** and **77** can be reduced. This makes it possible to suppress a change in quality or the like of ink left in the first and second heaters **76** and **77**. This enables reduction of an amount of waste ink having been changed in quality or the like and an amount of waste paper caused by a printing failure in the continuous paper **10**. Further, in the present preferred embodiment, the circulation duration ΔT (s) for which ink circulation is continued is determined on the basis of the duty values of the first and second heaters **76** and **77** and a flow rate of ink, and therefore the circulation duration ΔT (s) can be minimized.

[0078] Moreover, as is clear from the expression (1), the circulation duration ΔT (s) is proportional to the value $D1$ (%) that is the larger of the respective duty values of the first heater **76** and the second heater **77** for the above-described predetermined time period. That is, the circulation duration ΔT (s) is proportional to the duty values of the first and second heaters **76** and **77** for the predetermined time period. This is because large duty values of the first and second heaters **76** and **77** for the above-described predetermined time period are considered to mean generation of much heat and high temperatures of the heating element in the first and second heaters **76** and **77** and it is thus required to continue ink circulation for an accordingly prolonged time period.

[0079] Further, the control unit **9** substitutes the value $D1$ (%) that is the larger of the duty values of the first heater **76** and the second heater **77** for the above-described predetermined time period, into the expression (1). Thus, even in a case in which one of the heating elements in the first heater **76** and the second heater **77** generates extremely much heat and the temperature is high, the temperature of the one heating element can be sufficiently reduced. Consequently, a change in quality or the like of ink left near the one heating element can be sufficiently reduced.

[0080] Meanwhile, as is clear from the expression (1), the circulation duration ΔT (s) is inversely proportional to the flow rate Q_i (ml/min) of ink in the circulation path **41** for the above-described predetermined time period. This is because the higher the flow rate Q_i (ml/min) of ink is, the more ink can be brought close to the heating elements in the first and second heaters **76** and **77** during ink circulation and the sooner the temperatures of the heating elements in the first and second heaters **76** and **77** can be reduced. Consequently, the temperatures of the first and second heaters **76** and **77** can be efficiently reduced, which shortens the circulation duration ΔT (s) and improves the workability.

[0081] Subsequently, power supply to the inkjet printing apparatus **1** is resumed, and the control unit **9** drives the circulation pump **71** again in performing conveyance of the continuous paper **10**, printing on the continuous paper **10**, and ink circulation. Further, the control unit **9** performs conveyance of the continuous paper **10**, printing on the continuous paper **10**, and ink circulation while resuming driving the first and second heaters **76** and **77**.

[0082] As shown in FIG. **4**, after that, the control unit **9** determines whether to end conveyance of the continuous paper **10** and printing (step S5). The control unit **9** continues conveyance of the continuous paper **10** and printing when there remains image data to be printed (step S5: NO). After a while, when there is no image data to be printed (step S5: YES), the control unit **9** stops operations of each component and ends conveyance of the continuous paper **10**, printing on the continuous paper **10**, and ink circulation.

<3. Modifications>

[0083] Hereinabove, the preferred embodiment of the present invention has been described, but the present invention is not limited to the above-described preferred embodiment.

[0084] Further, the respective elements described in the above-described preferred embodiment may be appropriately combined unless contradiction occurs.

[0085] While the invention has been shown and described in detail, the foregoing description is in all aspects illustrative and not restrictive. It is therefore understood that numerous modifications and variations can be devised without departing from the scope of the invention.

Claims

1. An inkjet printing apparatus that discharges ink onto a printing medium to perform printing, comprising: an ink circulation path including a discharge head configured to discharge ink, a supply tank in which ink supplied to the discharge head is stored, a collecting tank in which ink collected from the discharge head is stored, and a feedback pipe connecting the collecting tank and the supply tank; a circulation pump that is interposed in the feedback pipe and is configured to deliver ink from the collecting tank to the supply tank via the feedback pipe; a heater that is interposed in the feedback pipe and is configured to heat ink flowing from the collecting tank to the supply tank; an input unit configured to receive a command for stopping driving the circulation pump; and a control unit electrically connected to each of the circulation pump, the heater, and the input unit, wherein the control unit is capable of sequentially performing a step (a) of, upon receipt of a signal regarding the command from the input unit, stopping driving the heater, and determining a circulation duration for which the circulation pump is kept operating and ink circulation in the circulation path is continued from a time of receipt of the signal, on the basis of a duty value of the heater for a predetermined time period before the time of receipt of the signal and a flow rate of ink in the circulation path for the predetermined time period, and a step (b) of stopping driving the circulation pump after elapse of the circulation duration from the time of receipt of the signal.
2. The inkjet printing apparatus according to claim 1, wherein the circulation duration is inversely proportional to the flow rate of ink in the circulation path for the predetermined time period.

3. The inkjet printing apparatus according to claim 1, wherein the circulation duration is proportional to the duty value of the heater for the predetermined time period.
